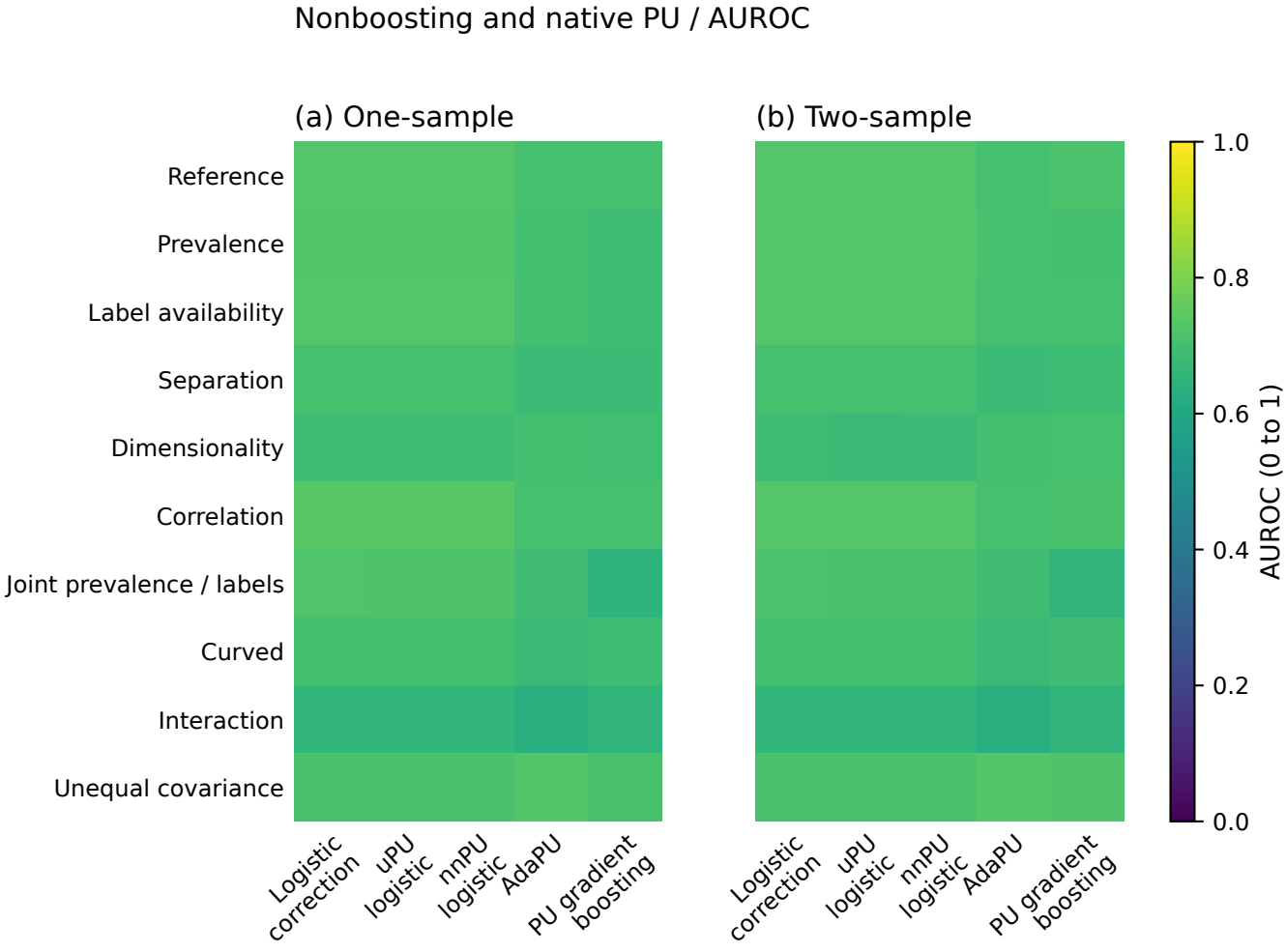
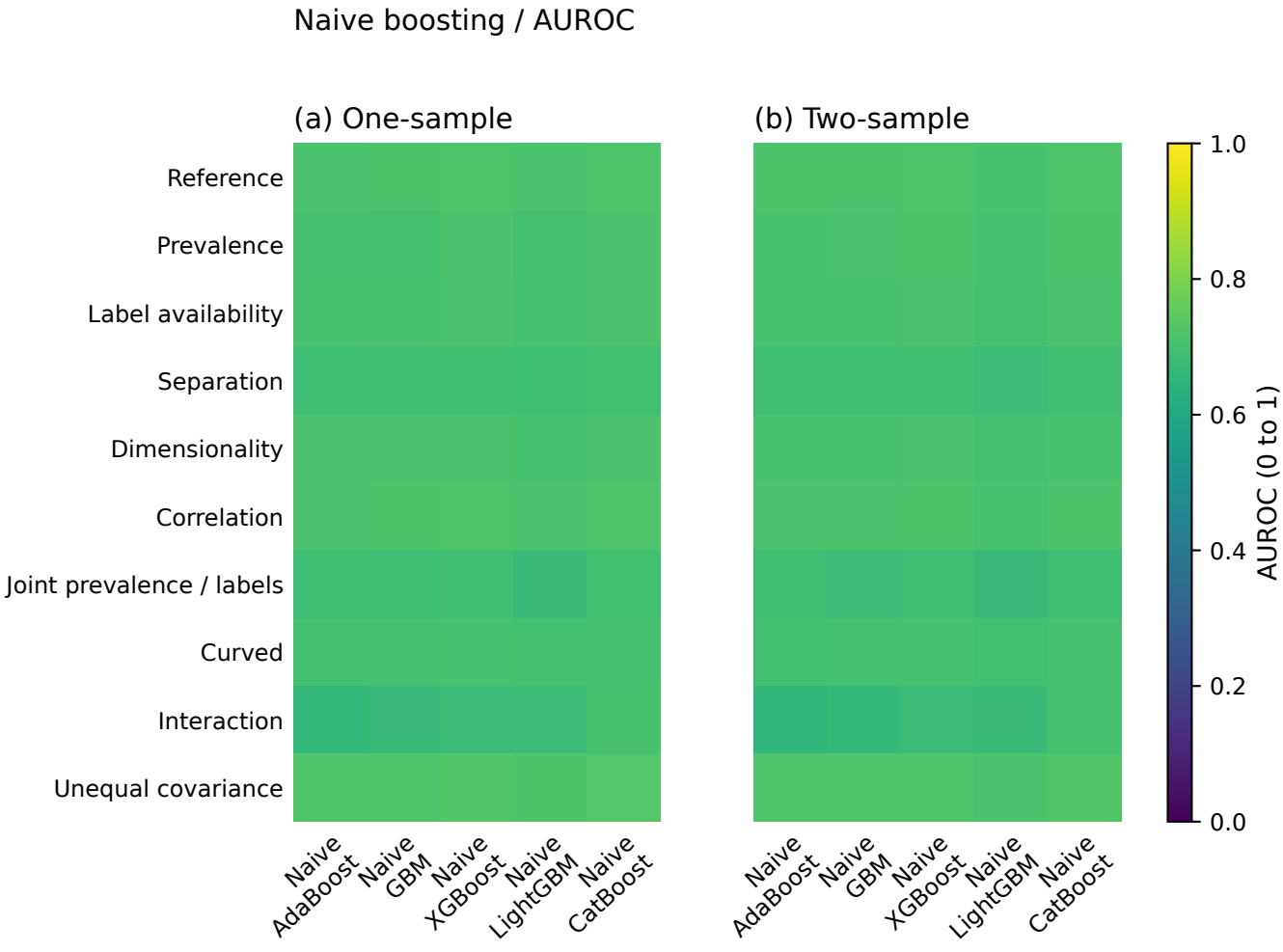
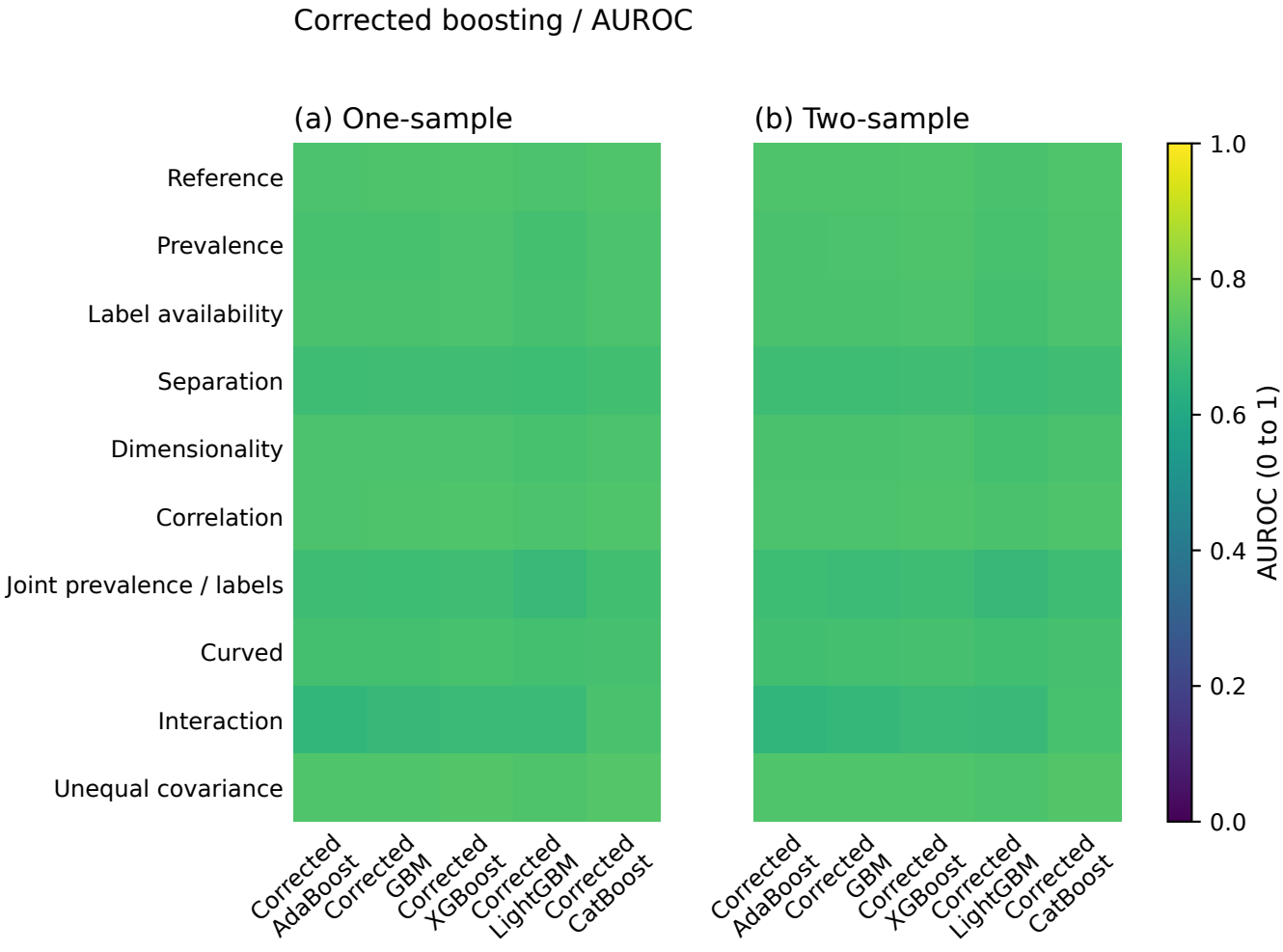
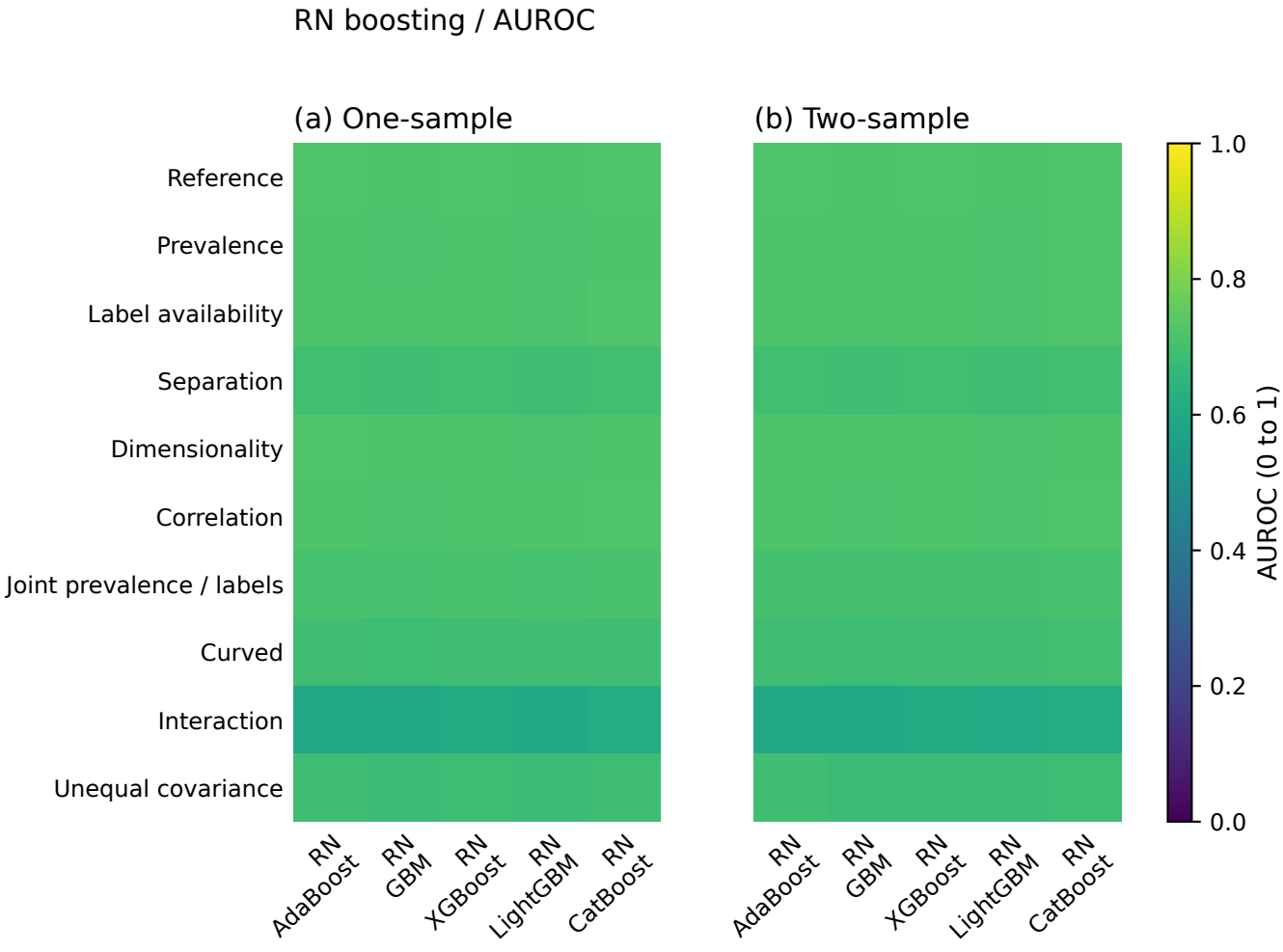
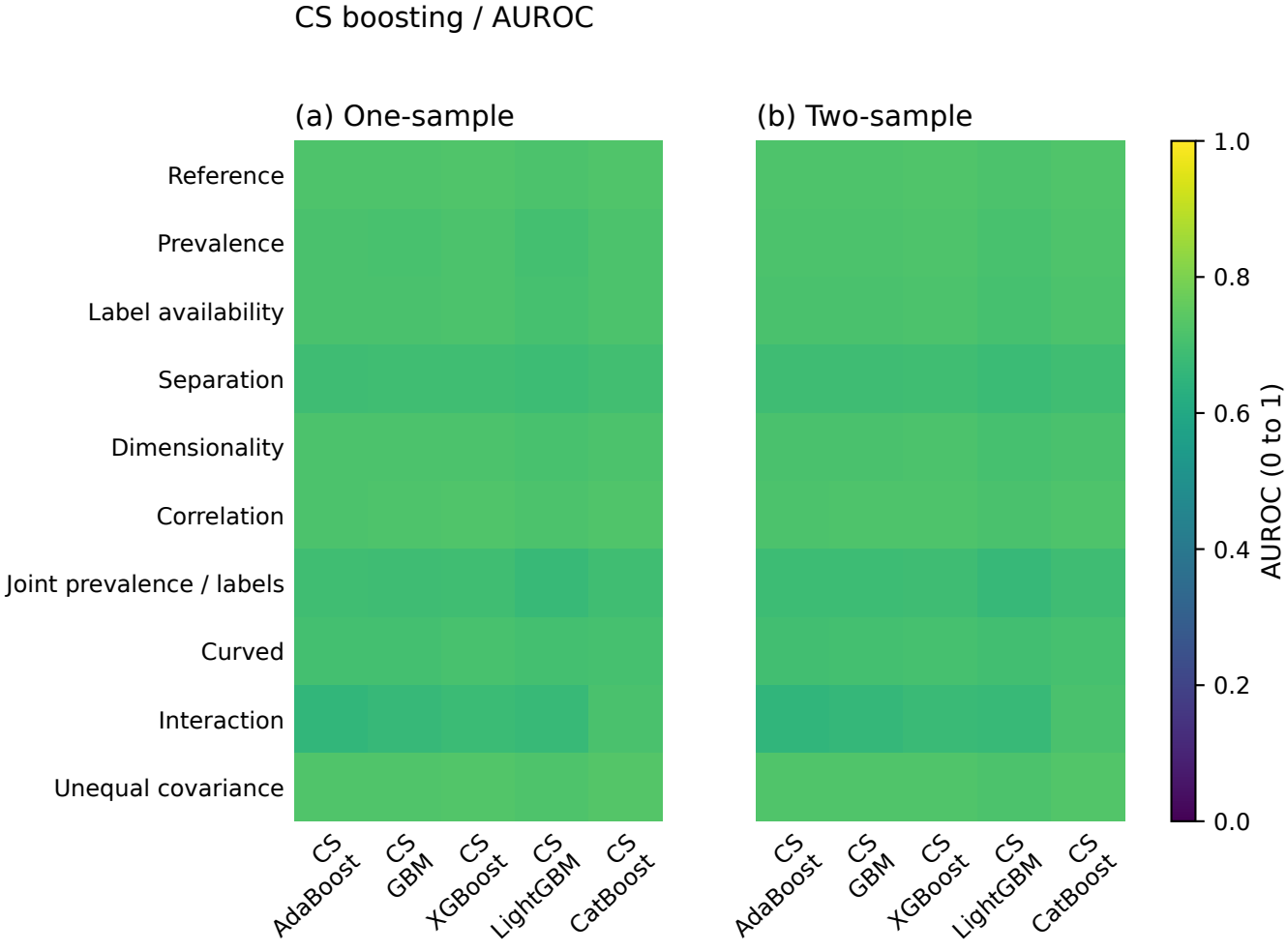
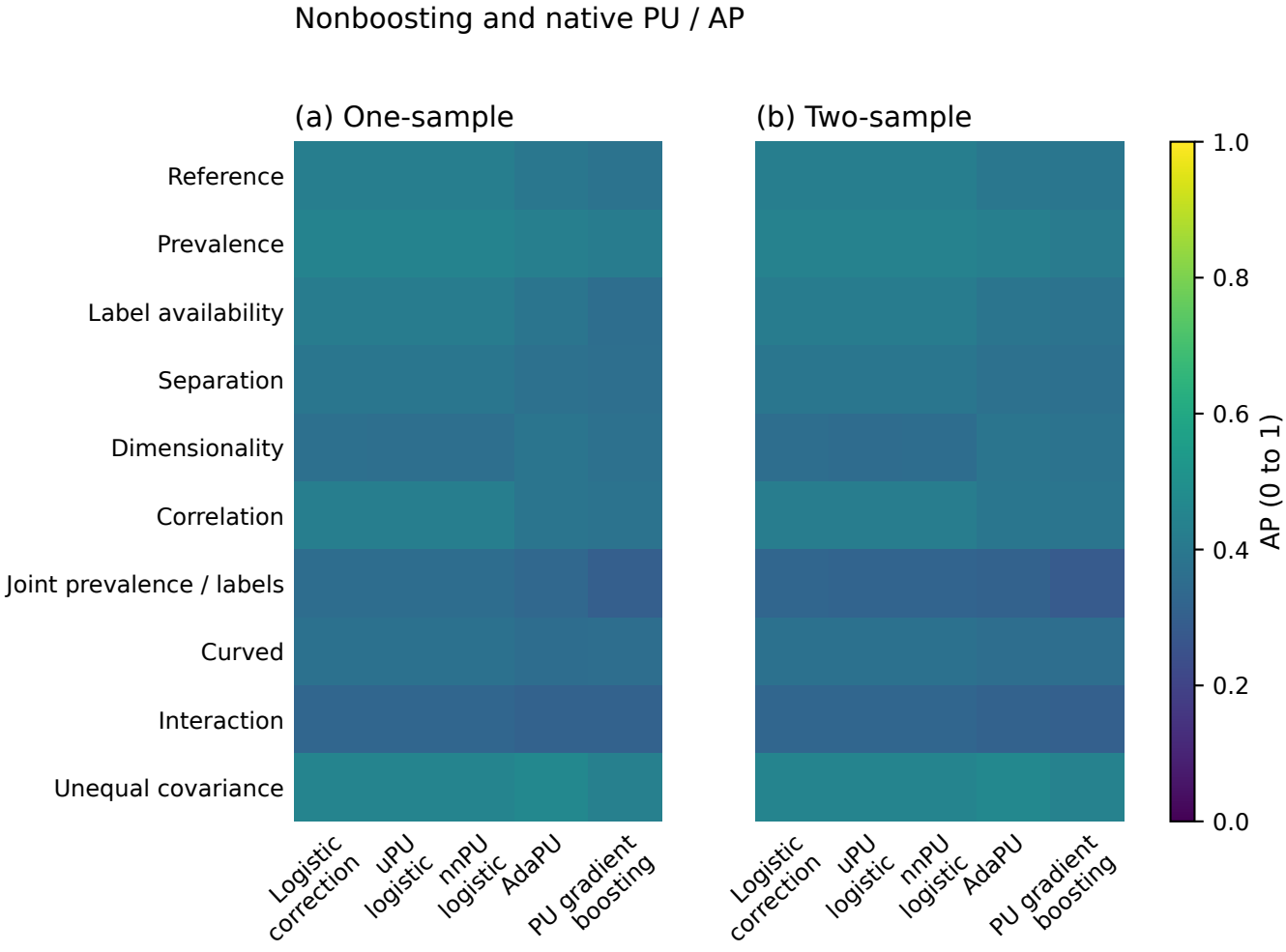

Nonboosting and native PU / AUROC
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

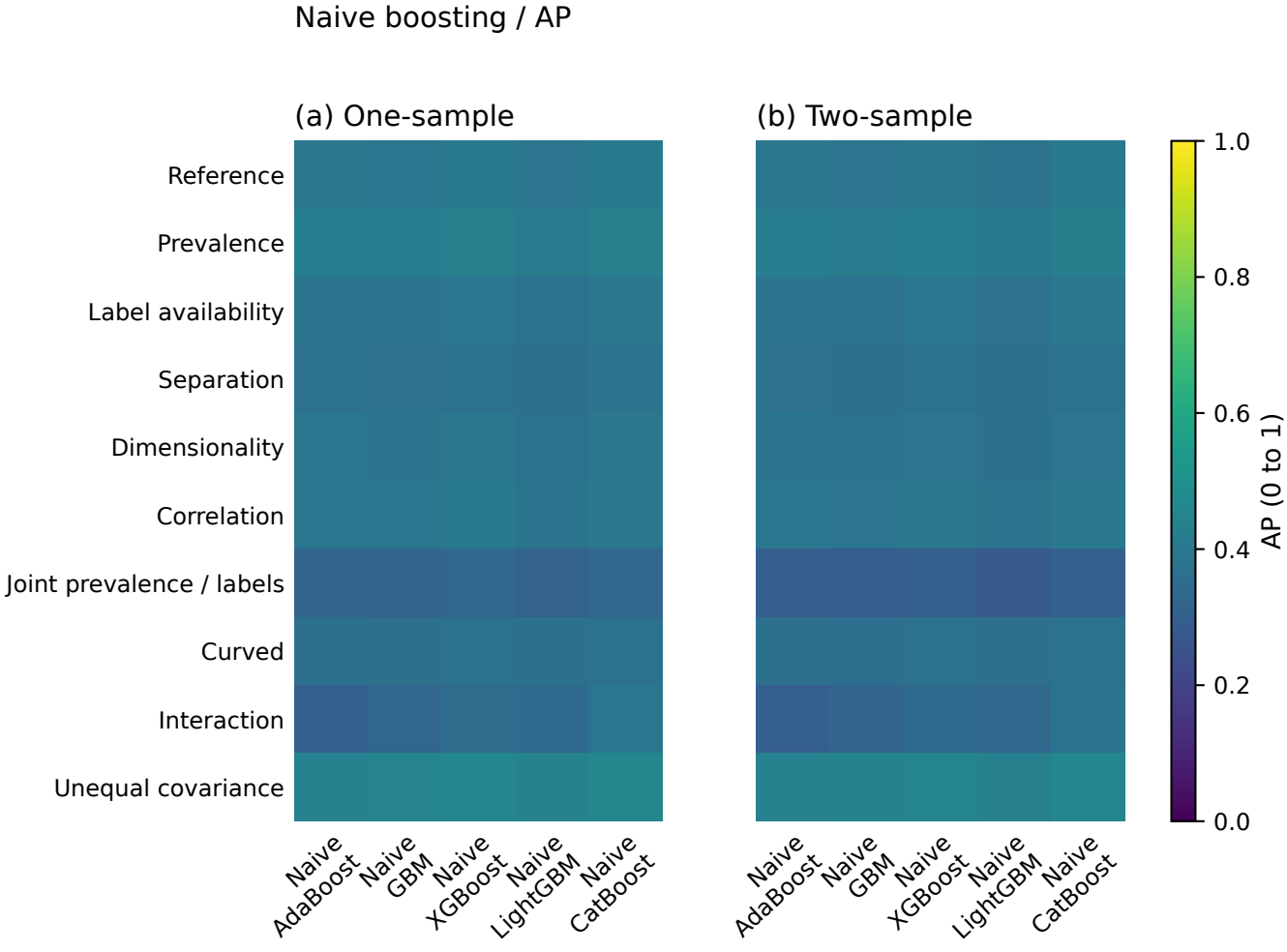

Naive boosting / AUROC
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

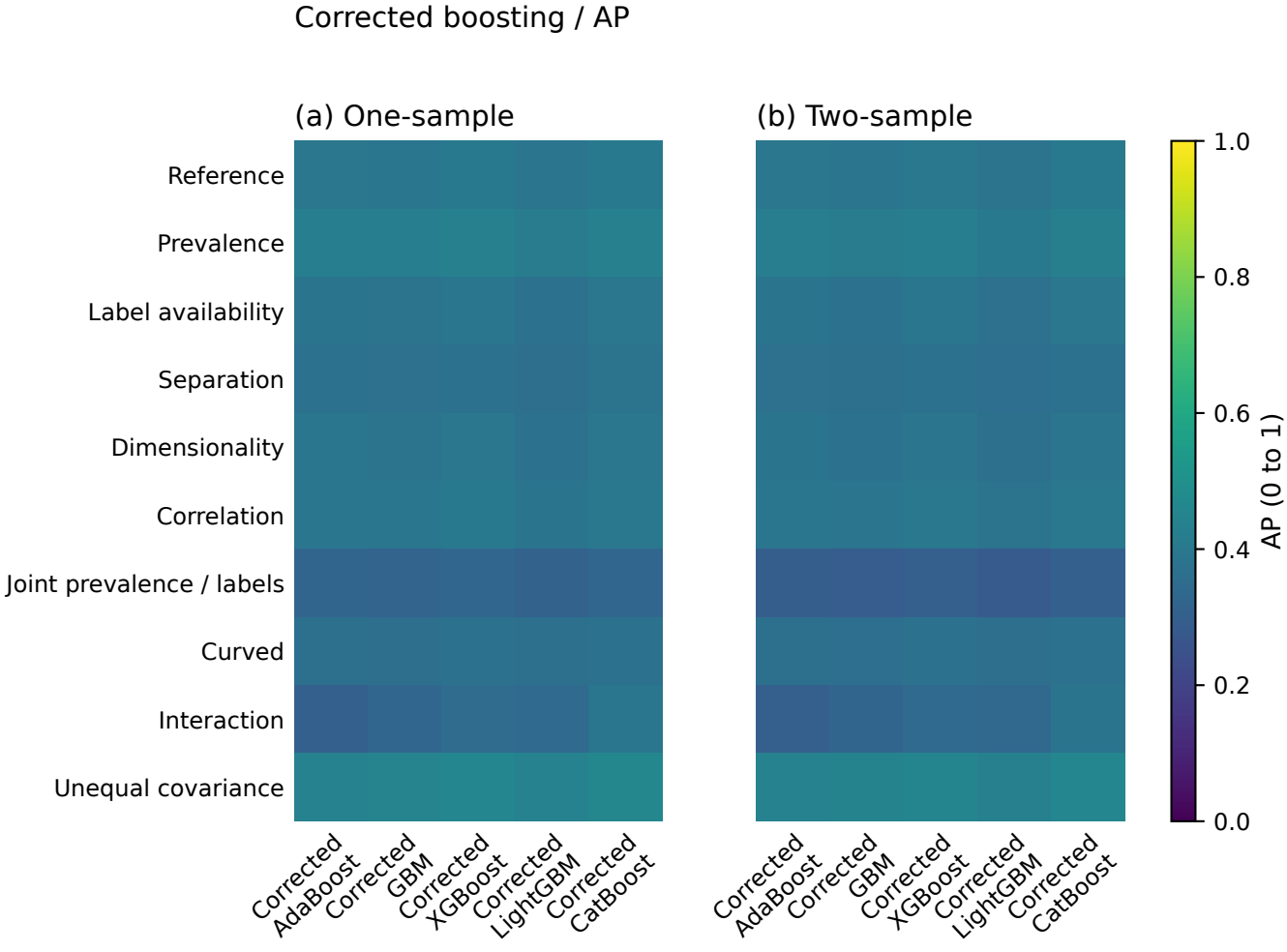

Corrected boosting / AUROC
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
Corrected AdaBoost
Corrected GBM
Corrected XGBoost
Corrected LightGBM
Corrected CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

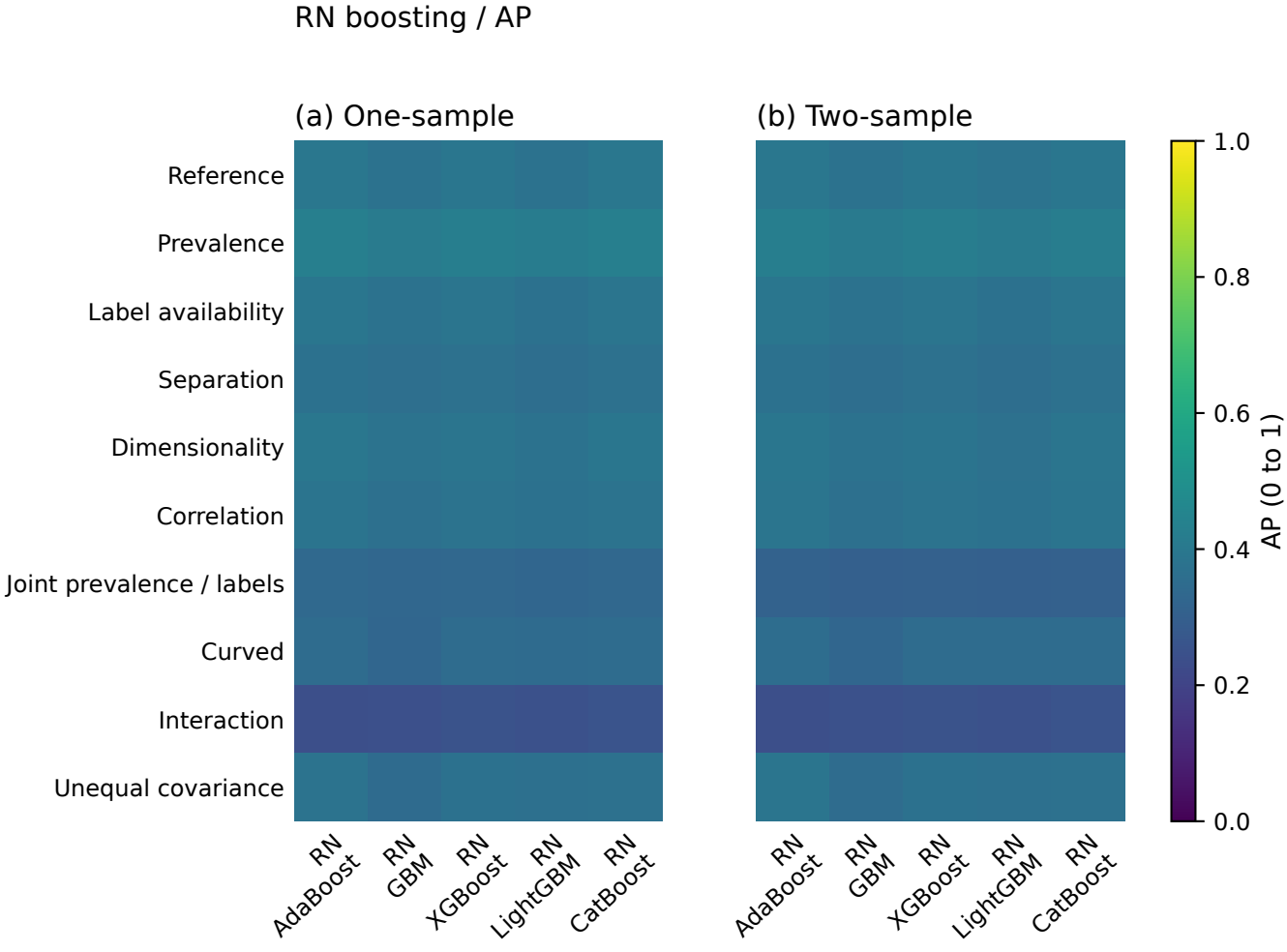

RN boosting / AUROC
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
RN AdaBoost
RN GBM
RN XGBoost
RN LightGBM
RN CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

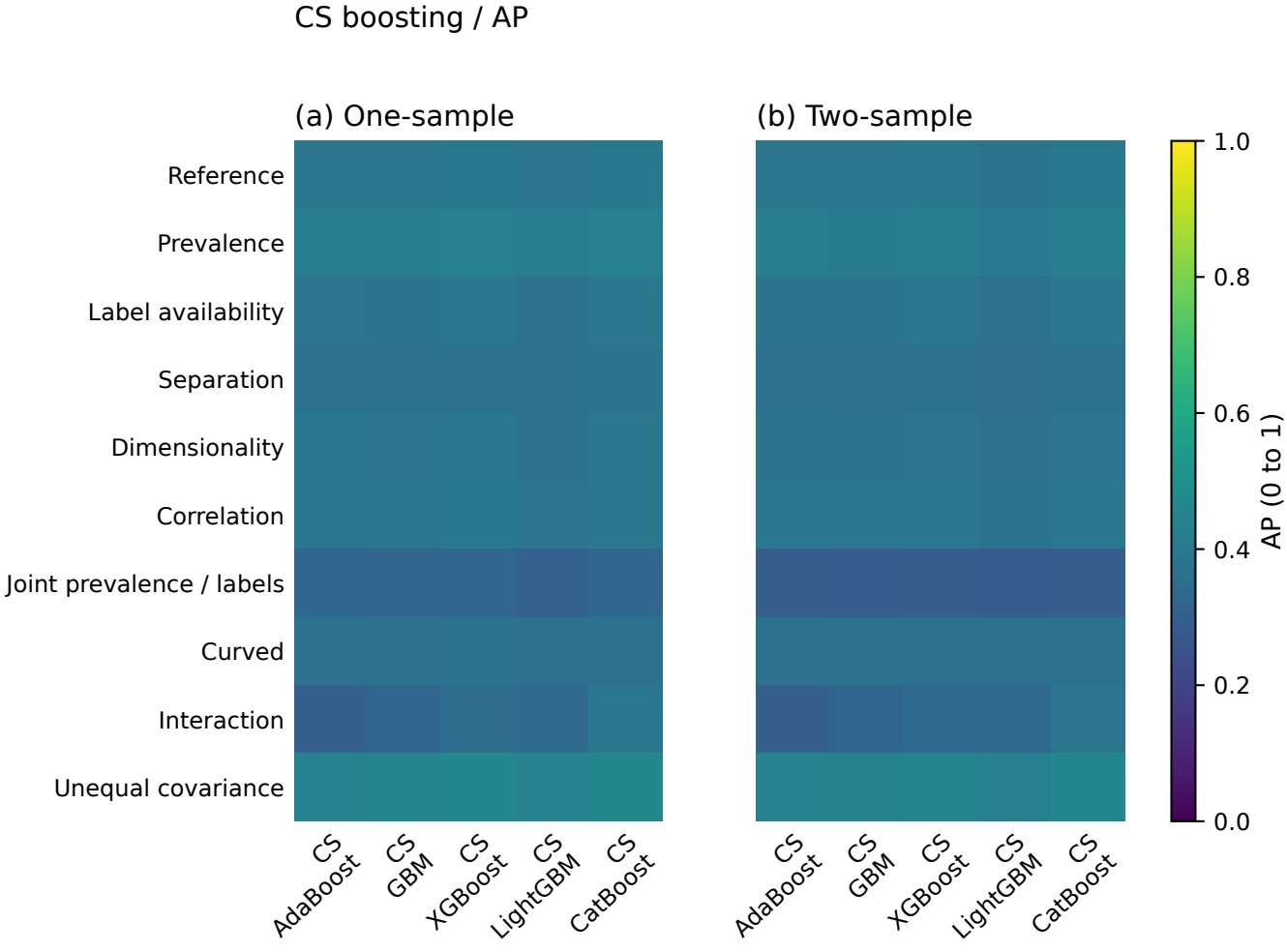

CS boosting / AUROC
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
CS AdaBoost
CS GBM
CS XGBoost
CS LightGBM
CS CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Nonboosting and native PU / AP
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Naive boosting / AP
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Corrected boosting / AP
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
Corrected AdaBoost
Corrected GBM
Corrected XGBoost
Corrected LightGBM
Corrected CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


RN boosting / AP
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
RN AdaBoost
RN GBM
RN XGBoost
RN LightGBM
RN CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


CS boosting / AP
(a) One-sample
(b) Two-sample
Reference
Prevalence
Label availability
Separation
Dimensionality
Correlation
Joint prevalence / labels
Curved
Interaction
Unequal covariance
CS AdaBoost
CS GBM
CS XGBoost
CS LightGBM
CS CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0